

A counterexample to the indexed interlacing claim in TLMC conjecture 00000007681

Local AI-assisted verification package

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Abstract

The first assertion of TLMC conjecture 00000007681 is false for the standard continuous q -Hermite polynomials. At $q = 3/4$, $n = 2$, and $k = 1$, the stated inequality requires $0 < -1/4$. All indices lie in their stated root lists. We derive the polynomials from their recurrence and give their complete, strictly increasing lists of real roots. The accompanying Lean file formalizes this bridge as well as the contradiction.

1 Statement and interpretation

The frozen source is conjecture 00000007681 at organizer commit 95acb520ec5607c826b8a997b1ef2fc82d6f7c57 [1]. Both language versions give the recurrence

$$2xH_n(x \mid q) = H_{n+1}(x \mid q) + (1 - q^n)H_{n-1}(x \mid q)$$

and order the zeros as $x_{n,1} < \dots < x_{n,n}$. The first asserted property is

$$x_{n+1,2k} < x_{n,k} < x_{n+1,2k+1} \quad (0 < q < 1). \quad (1)$$

We evaluate this only when all three indices exist.

The source does not list initial values. We use the standard continuous q -Hermite normalization $H_0 = 1$, $H_1 = 2x$. These initial values follow from the defining finite sum in NIST DLMF equation 18.28.16 [2]: for $n = 0$ the sum is 1, and for $n = 1$ it is $e^{i\theta} + e^{-i\theta} = 2\cos\theta$. The paper's use of this published normalization is explicit; the Lean file starts from these initial polynomials and the displayed recurrence.

2 Exact counterexample

Proposition 1. *Equation (1) is false for $q = 3/4$, $n = 2$, $k = 1$.*

Proof. The recurrence and initial values give

$$H_2(x \mid q) = 4x^2 - (1 - q), \quad H_3(x \mid q) = 8x^3 - 2x(2 - q - q^2).$$

At $q = 3/4$ these become

$$H_2(x \mid 3/4) = 4x^2 - \frac{1}{4} = 4 \left(x + \frac{1}{4} \right) \left(x - \frac{1}{4} \right),$$
$$H_3(x \mid 3/4) = x \left(8x^2 - \frac{11}{8} \right) = 8x \left(x + \frac{\sqrt{11}}{8} \right) \left(x - \frac{\sqrt{11}}{8} \right).$$

Their exhaustive increasing lists of real zeros are therefore

$$(x_{2,1}, x_{2,2}) = \left(-\frac{1}{4}, \frac{1}{4}\right), \quad (x_{3,1}, x_{3,2}, x_{3,3}) = \left(-\frac{\sqrt{11}}{8}, 0, \frac{\sqrt{11}}{8}\right).$$

The parameter satisfies $0 < 3/4 < 1$. The choice $n = 2, k = 1$ uses the first zero of H_2 and the second and third zeros of H_3 , so no out-of-range index is involved. The left-hand inequality in (1) is now $x_{3,2} < x_{2,1}$, or $0 < -1/4$, a contradiction. $\square$

3 Formal coverage and limits

The accompanying Lean source is `Results/Counterexample7681.lean`. It defines the polynomial family over $\mathbb{R}$, verifies the recurrence and the displayed evaluations, characterizes all real roots by if-and-only-if statements, and proves that the two root tables are strictly increasing and exhaustive. The predicate `ConjecturedInterlacing` translates (1) using zero-based `Fin` indices: source index k corresponds to the value $k - 1$. The theorem `conjectured_interlacing_false` negates that predicate.

The source also asserts a monotonicity and limiting property of root differences. We do not formalize that second assertion independently. The first assertion is necessary for the stated conjunction, so its counterexample refutes the conjunction regardless of the second assertion's precise interpretation.

This result diagnoses an indexing defect in the generated statement. It is not a new general theorem about orthogonal polynomials and does not refute the correctly indexed ordinary interlacing theorem. A bounded check of all 100 organizer pull-request titles and bodies, captured at 2026-09-21 01:13:39 UTC+08, found no matching conjecture number. This is not a claim of global priority. Build, axiom-audit, and replay results are recorded in the accompanying verification logs; organizer acceptance is not implied by local checks.

References

- [1] The Last Math Competition, conjecture 00000007681, frozen at commit 95acb520ec5607c826b8a997b1ef2fc82d6f7c57.
<https://github.com/The-Last-Math-Competition/The-Last-Math-Competition/blob/95acb520ec5607c826b8a997b1ef2fc82d6f7c57/conjectures/00000007681.md>.
- [2] NIST Digital Library of Mathematical Functions, Section 18.28(vi), equation 18.28.16, Continuous q -Hermite Polynomials. <https://dlmf.nist.gov/18.28.E16>. Accessed 21 September 2026.